Sanaul Haque

Irene_Chiandetti_Proposal

Assignment 1066

Document Details

Submission ID
trn:oid:::3117:637866153

Submission Date
Oct 12, 2026, 10:10 PM GMT+6

Download Date
Oct 12, 2026, 10:13 PM GMT+6

File Name
Irene_Chiandetti_Proposal.pdf

File Size
408.7 KB

7 Pages

3,273 Words

20,549 Characters

30% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

26 Not Cited or Quoted 20%
Matches with neither in-text citation nor quotation marks

11 Missing Quotations 3%
Matches that are still very similar to source material

8 Missing Citation 7%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

26% Internet sources
24% Publications
26% Submitted works (Student Papers)

Match Groups

- **26** Not Cited or Quoted 20%
  Matches with neither in-text citation nor quotation marks
- **11** Missing Quotations 3%
  Matches that are still very similar to source material
- **8** Missing Citation 7%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

26% Internet sources
24% Publications
26% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	carleton.scholaris.ca 2%
2	Student papers	De La Salle University - Manila on 2026-08-10 2%
3	Internet	sinbad2.ujaen.es 1%
4	Student papers	Galatasaray University on 2026-10-09 1%
5	Student papers	Khalifa University of Science Technology and Research on 2025-04-27 1%
6	Internet	www.preprints.org 1%
7	Internet	jurnal.iaii.or.id 1%
8	Student papers	Roehampton University on 2025-05-27 1%
9	Internet	scholarspace.manoa.hawaii.edu 1%
10	Student papers	General Sir John Kotelawala Defence University on 2023-11-10 1%

11	Internet	
arxiv.org		1%
12	Internet	
assets-eu.researchsquare.com		<1%
13	Student papers	
Canadian University of Dubai on 2026-07-05		<1%
14	Student papers	
Oklahoma State University on 2022-12-06		<1%
15	Internet	
stax.strath.ac.uk		<1%
16	Student papers	
University of Massachusetts - Amherst on 2024-12-11		<1%
17	Internet	
www.medrxiv.org		<1%
18	Internet	
www.researchsquare.com		<1%
19	Internet	
www.cambridge.org		<1%
20	Internet	
thesis.cust.edu.pk		<1%
21	Publication	
Muhammad Oka Bagus Wibowo, Syukron Abu Ishaq Alfarozi, Sunu Wibirama. "Pr...		<1%
22	Internet	
www.hindawi.com		<1%
23	Student papers	
Birkbeck College on 2025-06-02		<1%
24	Internet	
assets.researchsquare.com		<1%

25 Internet
en.wikipedia.org <1%

26 Internet
public-pages-files-2025.frontiersin.org <1%

27 Student papers
University of Western Australia on 2026-06-25 <1%

28 Publication
Chenxi Pan, Jing Ma, Xiao Yuan, Renren Li et al. "Atrophy of specific amygdala sub... <1%

29 Student papers
University of Newcastle upon Tyne on 2025-08-29 <1%

30 Internet
bighealth.fudan.edu.cn <1%

31 Internet
www.twc.edu.hk <1%

32 Student papers
Murdoch University on 2026-09-10 <1%

33 Publication
Xiang Yang, Shuo Gao, Peichang Wang. "TUBB4B mRNA in CSF and plasma exoso... <1%

34 Internet
bradscholars.brad.ac.uk <1%

35 Internet
5dok.org <1%

36 Student papers
Gitam University on 2026-08-06 <1%

37 Student papers
Tilburg University on 2026-05-23 <1%

38 Publication
Wei Si, XiaoHan Ma, Jiayue Lin, Ting Xu, Rui Wang, Aiyong Zhu, Wenting Cao. "Dev... <1%

39 Internet

mdpi-res.com <1%

40 Internet

www.mdpi.com <1%

EYE-MCI: A Brief Multidomain Virtual-Reality Eye-Tracking and Machine-Learning Tool for Screening Mild Cognitive Impairment

Research Problem

Alzheimer's disease (AD) is a neurodegenerative disease and the most prevalent cause of dementia, accounting for around 60-80% of cases [1]. The progression of AD begins with an initial amyloid plaque formation and subtle cognitive decline, preceding the cascade of pathological events that culminate with network dysfunctions and injury and degeneration of neurons [1], [2]. Neurodegeneration becomes irreversible in the later stages of the disease, but early anti-amyloid treatment can delay the onset of symptoms [3], [4]. Mild cognitive impairment (MCI) is therefore an important clinical stage for early identification, monitoring and referral. Current brief screens such as the Montreal Cognitive Assessment (MoCA) are inexpensive and useful, but performance is influenced by education, language, examiner administration, and testing context, giving rise to cultural and educational biases that are hard to overcome [5]. More definitive amyloid and tau assessments can identify AD-related pathology, including positron-emission tomography, cerebrospinal-fluid testing or blood-biomarkers [6]. Nevertheless, the former approaches remain expensive or invasive, while blood biomarkers are still being integrated into specialists pathway [6], [7].

Eye movements provide continuous, objective measures of top-down, goal-directed behavior and spatial attention within ecologically valid environments [8]. However, most existing studies do not exploit the full potential of eye-tracking as a diagnostic tool, as they examine one paradigm, use small samples, or classify participants against a cognitive-test threshold rather than an independent clinical diagnosis [9], [10]. Xu et al. developed a five-minute VR Eye-tracking Cognitive Assessment (VECA) that predicted MoCA scores with strong correlation, but its principal endpoint was MoCA-defined impairment: it did not independently establish MCI or AD pathology [11]. The purpose of this project is to propose EYE-MCI, a short VR battery that combines established eye-movement paradigms, validates the tool against clinically adjudicated MCI, and explores whether its gaze profile is associated with AD-related biomarkers, to address the gaps present in the existing literature.

Existing Literature

Systematic reviews show that MCI is associated with abnormalities across both saccadic and cognitively informed eye-tracking tasks, but no single measure captures the heterogeneity of MCI [12], [13]. Antisaccade tasks are particularly relevant because they require suppression of a reflexive response and generation of a voluntary saccade [10]; MCI studies report higher directional-error rates and poorer error correction, indicating early executive-control impairment [13–15]. Prosaccades are less specific but provide a baseline for visual orienting and motor latency [14].

Visual paired-comparison (VPC) tasks assess recognition memory through novelty preference [14], [16]. A 2024 meta-analysis of 12 studies and 1,022 participants reported pooled sensitivity of 0.84 and specificity of 0.75, supporting VPC as one of the most replicated eye-tracking approaches to early memory decline [16]. Scene-memory paradigms such as VisMET measure preferential viewing of a changed or manipulated region and have distinguished memory-impaired participants in less than five minutes [17]. Visual-search studies also show that participants with MCI make more fixations, inspect distractors longer, and search less efficiently than healthy controls [18].

Rapid multidomain batteries demonstrate feasibility. Oyama et al. assessed several cognitive domains in approximately three minutes [19], while Xu et al. integrated VR, gaze-based responses, and machine learning in approximately five minutes [11]. Nevertheless, many such batteries predominantly measure

how long a participant looks at the correct answer rather than process-level features such as inhibitory errors, novelty preference, or memory-guided gaze. Recent reviews of digital AD biomarkers also identify limited external validation and rare calibration analysis despite encouraging receiver operating characteristic area under the curve (ROC-AUC) values – a parameter often used to evaluate classification performance in early-stage diagnostic tools [20], [21]. EYE-MCI therefore aims to combine rapid administration with theoretically selected, interpretable gaze variables and explicit model-calibration assessment.

Research Questions

This study aims to develop and preliminarily validate EYE-MCI in adults aged 60–85 years. It will answer:

1. Are there significant differences between clinically identified MCI and cognitively healthy controls in: (a) prosaccade median latency and endpoint accuracy; (b) antisaccade directional-error rate, uncorrected-error rate, and correct-response latency; (c) VPC novelty-preference score; (d) virtual scene-change accuracy and proportion of dwell time on the changed region; (e) visual-search completion time, target accuracy, and distractor dwell time?
2. How accurately does a model combining these variables distinguish MCI from healthy cognition, measured by ROC-AUC, sensitivity, specificity, positive and negative predictive values, classification accuracy, and calibration?
3. Which tasks and variables contribute most strongly to classification, and does the multidomain battery outperform each task alone?
4. How strongly are the overall and task-specific eye-tracking scores associated with MoCA and domain-specific neuropsychological performance?
5. Does eye tracking improve MCI classification beyond age, sex, education, and MoCA?
6. Among participants meeting clinical criteria for MCI, are EYE-MCI scores and memory-related gaze measures associated with established biomarkers of Alzheimer-related pathology?

Methodology

Design and sample A cross-sectional, blinded diagnostic-accuracy study will recruit approximately 160 participants: 80 with MCI and 80 cognitively healthy controls. Adults aged 60–85 will be included because this range captures clinically relevant late-onset MCI while limiting heterogeneity from young-onset disorders and extreme sensory or frailty-related barriers. MCI will be identified independently of EYE-MCI using established criteria: evidence of cognitive decline, objective impairment in one or more cognitive domains, largely preserved independence in everyday activities, and absence of dementia [24]. Objective impairment will be assessed across episodic memory, executive/attentional, language, and visuospatial domains using standardized measures such as auditory-verbal learning and delayed recall, Trail Making Tests A–B, verbal fluency, and Rey–Osterrieth Complex Figure [25]. MoCA will be administered as a global cognitive comparator. Exclusion criteria will be delirium, major unstable psychiatric or neurological illness, severe uncorrected visual impairment, inability to understand instructions, or failure to obtain interpretable calibration. Visual acuity, ophthalmic disease, depressive symptoms, medication, language, and education will be documented as potential confounders.

Biomarker assessment Within the MCI group, Alzheimer-related biomarkers will be analysed separately to test whether EYE-MCI abnormalities are associated with underlying AD pathology. Biomarker data will only be included in the analysis if they can be ethically or practically obtained; unavailable data will be transparently reported. Where available, AD biomarkers will include plasma p-tau217, CSF Aβ42/Aβ40 and phosphorylated tau, and amyloid or tau PET [6], [7], [25]. These will provide a secondary biological

validation of EYE-MCI, testing whether the tool captures biologically relevant cognitive impairment beyond its association with MoCA scores.

VR and software development A seated head-mounted display with integrated binocular eye tracking of approximately 90 Hz or higher will be used, providing sufficient temporal resolution for the saccadic latency measures included in the battery. Following the portable VR–eye-tracking framework demonstrated by Xu et al. [11], the EYE-MCI application will present the cognitive tasks, record gaze responses, and integrate these data for subsequent machine-learning analysis. The application will be developed in Unity using C#, an architecture previously used in immersive cognitive-assessment systems [22]. The software will control stimulus presentation and task randomization, perform calibration and periodic calibration-quality checks, and synchronously record gaze and task-event timestamps. Raw or minimally processed gaze data will be retained where available to permit quality control and later reanalysis. A fixed software version and prespecified preprocessing criteria will be used throughout participant testing to improve reproducibility and reduce analysis bias. Reporting of the diagnostic model and its evaluation will follow STARD-AI principles [23].

Stimuli and battery Stimuli will be prepared for older adults using high contrast, large objects, controlled luminance, limited text, and standardized angular size and eccentricity. Two matched forms will be produced for reliability and future retesting. A pilot with healthy older adults will identify unclear instructions, ceiling or floor effects, visual-recognition problems, and cybersickness. The initial battery will contain: (1) 12–16 prosaccade trials as a baseline; (2) 20–24 antisaccade trials to estimate errors reliably; (3) 6–8 VPC familiarization-test pairs; (4) six naturalistic VR rooms in which an object is moved, removed, replaced, or unchanged; and (5) 8–12 memory-guided visual-search trials with two difficulty levels. Stimulus time is expected to be 5–7 minutes, with 7–10 minutes of total chair time including calibration and instructions. These durations are design estimates informed by rapid batteries [11], [19] and the under-five-minute VisMET paradigm [17]; pilot reliability, rather than convenience alone, will determine whether tasks can be shortened.

Eye-tracking variables and quality control Primary variables will be prosaccade latency; antisaccade directional and uncorrected-error rates; VPC novelty preference; scene-change accuracy and changed-region dwell proportion; and visual-search time, target accuracy, and distractor dwell. Secondary variables will include first-target fixation latency, revisits, endpoint error, and within-person variability. Calibration will be validated centrally and peripherally, with drift checks before each major task. Prespecified thresholds will define valid samples and trials. Missing gaze data will not be coded as cognitive errors. Calibration error, percentage of valid samples, and recalibration frequency will be used to assess data interpretability and in sensitivity analyses.

Acceptability and feasibility The System Usability Scale will measure usability, while a brief pre/post cybersickness questionnaire will record nausea, dizziness, eyestrain, headache, and disorientation. Completion rate, calibration time, total administration time, assistance required, repeated instructions, recalibration, and unusable-data rate will be recorded, to ensure that EYE-MCI allows older adults to comfortably complete the assessment.

Procedure

After consent and eligibility screening, participants will complete visual and clinical background measures, MoCA, neuropsychological testing, and functional assessment. The assessor assigning clinical status will remain blinded to EYE-MCI results. Participants will then complete headset fitting, calibration, practice trials, and the fixed-duration battery in a seated position. Task order will be partially counterbalanced, while delayed memory trials will return at the end of the battery. Usability and

cybersickness ratings will follow immediately. Where available, AD biomarkers will be analysed in an exploratory MCI subgroup as a secondary biological validation of EYE-MCI.

Data Analysis

Group differences in prespecified eye-tracking outcomes will be tested using regression models adjusted for age, sex, education, visual acuity, depressive symptoms, and calibration quality. Construct validity will be assessed by correlating memory, antisaccade, and visual-search measures with corresponding neuropsychological domains. Following Xu et al. [11], multiple supervised models will be compared using cross-validation; penalized logistic regression will be the primary classifier, with support-vector classification and gradient-boosted trees as secondary models. Participant-level folds will prevent data leakage, and all preprocessing and tuning will occur within training data only [23], [26]. Model performance will be assessed using ROC-AUC, sensitivity, specificity, predictive values, and calibration, while ablation analysis will identify which tasks add the most discriminatory value. Exploratory biomarker analyses will test associations between EYE-MCI scores and AD-related biomarker measures.

Expected Outcome

EYE-MCI is expected to identify a multidomain gaze profile in which MCI is associated with more antisaccade errors, lower novelty preference, poorer scene-change detection, and slower, distractor-heavy search. The combined model is expected to outperform any individual task and to add information beyond demographics and MoCA. Memory-related gaze measures may show the strongest association with AD-related biomarker abnormality, but this analysis will be interpreted as preliminary biological validity rather than diagnosis or prognosis, given the contradictory evidence on the relevance of biomarkers alone in diagnosis of the MCI population [27]. The main output will be a calibrated probability of MCI and a recommendation for further clinical assessment.

Potential Limitations

- The preliminary sample may be insufficient for complex machine learning, biomarker subgroup comparisons, or definitive subgroup fairness analysis.
- A case-control sample can overestimate performance if groups are too clinically distinct.
- Eye tracking is affected by ocular disease, fatigue, medication, headset fit, and digital familiarity.
- Biomarkers may be unavailable for many participants and different assays are not interchangeable.
- VR may increase naturalistic context but also adds cybersickness and interface demands.
- Cross-sectional associations cannot establish future conversion to AD.

These limitations will be addressed through conservative modelling, blinded clinical classification, transparent missing-data reporting, and external validation as a required next stage.

Conclusion

EYE-MCI extends Xu et al.'s VR eye-tracking framework [11] by validating performance against clinically defined MCI based on functional and multidomain neuropsychological assessment [24], rather than mainly against MoCA-derived impairment. MoCA will remain a comparator, while amyloid- and tau-related biomarkers will provide secondary biological validation by testing whether abnormal EYE-MCI performance is associated with underlying AD pathology [6], [7]. This allows the study to assess whether the tool identifies clinically meaningful MCI while also capturing biologically relevant AD-related changes.

Project Practicalities

The project is planned over a course of 12 months: *months 1-3*, software architecture, stimuli, ethics, and technical validation; *months 4-5*, older-adult pilot and revision; *months 6-9*, recruitment and data collection through a memory-clinic partner; *months 10-11*, preprocessing, statistical modelling, and reliability checks; *month 12*, reporting and dissemination. Essential resources include one VR headset with integrated eye tracking, a development computer, Unity/OpenXR software, secure storage, clinical and neuropsychological supervision, and access to a recruitment site. Biomarker analyses will primarily use clinically available data to contain cost and participant burden.

Post-program Plan

After internal validation, the locked model will be tested prospectively in a second clinic and in participants older than 85 or younger than 60. Future work will evaluate alternate-language forms and device-to-device portability. A longitudinal study will then test whether baseline gaze profiles predict cognitive and functional decline or progression from MCI to AD dementia. A final translational phase would compare a monitor-based version with immersive VR to determine whether ecological context provides measurable benefit beyond the headset itself.

Research Topic

This topic is timely and clinically urgent. Populations are ageing, memory services face increasing demand, and disease-modifying treatment pathways make early, accurate triage more consequential. Existing cognitive screens are accessible but imperfect, while definitive biomarker tests remain specialist resources. Eye tracking is non-invasive, quantitative, and increasingly portable; VR can add controlled everyday contexts rather than simply presenting paper-test questions on a headset. By integrating interpretable gaze measures, cautious machine learning, clinical diagnosis, usability testing, and exploratory biomarker validation, EYE-MCI could offer researchers and healthcare systems a scalable pathway between brief cognitive screening and expensive confirmatory assessment.

References

[1] Q. Jiang et al., "Antiageing strategy for neurodegenerative diseases: from mechanisms to clinical advances," *Signal Transduction and Targeted Therapy*, vol. 10, no. 1, Mar. 2025, doi: 10.1038/s41392-025-02145-7.

[2] K. G. Yiannopoulou and S. G. Papageorgiou, "Current and Future Treatments in Alzheimer Disease: An Update," *Journal of Central Nervous System Disease*, vol. 12, no. 1, p. 117957352090739, Feb. 2020, doi: 10.1177/1179573520907397.

[3] S. J. Blondell, R. Hammersley-Mather, and J. L. Veerman, "Does physical activity prevent cognitive decline and dementia?: A systematic review and meta-analysis of longitudinal studies," *BMC Public Health*, vol. 14, no. 1, May 2014.

[4] J. Viña and J. Sanz-Ros, "Alzheimer's disease: Only prevention makes sense," *European Journal of Clinical Investigation*, vol. 48, no. 10, pp. 1–1, Oct. 2018, doi: 10.1111/eci.13005.

[5] P. Hadinezhad and M. Noroozian, "Montreal Cognitive Assessment (MoCA) Scale: Strengths, Limitations, and Implication for Clinical Practice," *Iranian Journal of Psychiatry*, Dec. 2025, doi: 10.18502/ijps.v21i1.20571.

[6] C. R. Jack, "Revised Criteria for Diagnosis and Staging of Alzheimer's Disease," *Alzheimer's & Dementia*, vol. 20, no. S2, Dec. 2024, doi: 10.1002/alz.095054.

[7] S. Palmqvist et al., "Alzheimer's Association Clinical Practice Guideline on the use of blood-based biomarkers in the diagnostic workup of suspected Alzheimer's disease within specialized care settings," *Alzheimer's & Dementia*, vol. 21, no. 7, July 2025, doi: 10.1002/alz.70535.

[8] R. J. Molitor, P. C. Ko, and B. A. Ally, "Eye Movements in Alzheimer's Disease," *Journal of Alzheimer's disease* : JAD, vol. 44, no. 1, pp. 1–12, 2015, doi: 10.3233/JAD-141173.

[9] S.-I. Tokushige et al., "Early detection of cognitive decline in Alzheimer's disease using eye tracking," *Frontiers in aging neuroscience*, vol. 15. Switzerland, p. 1123456, Mar. 21, 2023. doi: 10.3389/fnagi.2023.1123456.

[10] N. Noiret et al., "Saccadic Eye Movements and Attentional Control in Alzheimer's Disease," *Archives of Clinical Neuropsychology*, vol. 33, no. 1, pp. 1–13, May 2017, doi: 10.1093/arclin/acx044.

[11] Y. Xu, C. Zhang, B. Pan, Q. Yuan, and X. Zhang, "A portable and efficient dementia screening tool using eye tracking machine learning and virtual reality," npj *Digital Medicine*, vol. 7, no. 1, Aug. 2024, doi: 10.1038/s41746-024-01206-5.

[12] A. Wolf, Kornkanok Tripanpitak, S. Umeda, and Mihoko Otake-Matsuura, "Eye-tracking paradigms for the assessment of mild cognitive impairment: a systematic review," *Frontiers in Psychology*, vol. 14, July 2023, doi: 10.3389/fpsyg.2023.1197567.

[13] T. D. W. Wilcockson *et al.*, "Abnormalities of saccadic eye movements in dementia due to Alzheimer's disease and mild cognitive impairment," *Aging*, vol. 11, no. 15, pp. 5389–5398, Aug. 2019, doi: 10.18632/aging.102118.

[14] G. L. Verón, G. E. Juantorena, G. Keller, L. Crivelli, and J. E. Kamienkowski, "Eye tracking as a diagnostic tool in Alzheimer's disease, mild cognitive impairment, and related dementias: a systematic

review," *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, vol. 18, no. 1, Dec. 2025, doi: 10.1002/dad2.70238.

[15] N. Chehrehnegar et al., "Behavioral and cognitive markers of mild cognitive impairment: diagnostic value of saccadic eye movements and Simon task," *Aging Clinical and Experimental Research*, Jan. 2019, doi: 10.1007/s40520-019-01121-w.

[16] M. A. Boujelbane *et al.*, "Eye Tracking During Visual Paired-Comparison Tasks: A Systematic Review and Meta-Analysis of the Diagnostic Test Accuracy for Detecting Cognitive Decline," *Journal of Alzheimer s Disease*, vol. 99, no. 1, pp. 207–221, Apr. 2024, doi: 10.3233/jad-240028.

[17] R. U. Haque et al., "VisMET: a passive, efficient, and sensitive assessment of visuospatial memory in healthy aging, mild cognitive impairment, and Alzheimer's disease," *Learning & Memory*, vol. 26, no. 3, pp. 93–100, Feb. 2019, doi: 10.1101/lm.048124.118.

[18] M. Luísa *et al.*, "Visual Search Efficiency in Mild Cognitive Impairment and Alzheimer's Disease: An Eye Movement Study," *Journal of Alzheimer s Disease*, vol. 75, no. 1, pp. 261–275, May 2020, doi: 10.3233/jad-190690.

[19] A. Oyama et al., "Novel Method for Rapid Assessment of Cognitive Impairment Using High-Performance Eye-Tracking Technology," Scientific Reports, vol. 9, no. 1, Sept. 2019, doi: 10.1038/s41598-019-49275-x.

[20] W. Qi et al., "Alzheimer's disease digital biomarkers multidimensional landscape and AI model scoping review," npj Digital Medicine, vol. 8, no. 1, June 2025, doi: 10.1038/s41746-025-01640-z.

[21] D. M. A. S. Elkahwagy, C. J. Kiriacos, and M. Mansour, "Logistic regression and other statistical tools in diagnostic biomarker studies," Clinical and Translational Oncology, vol. 26, no. 9, pp. 2172–2180, Mar. 2024, doi: 10.1007/s12094-024-03413-8.

[22] K. Ijaz, N. Ahmadpour, S. L. Naismith, and R. A. Calvo, "An immersive virtual reality platform for assessing spatial navigation memory in predementia screening: Feasibility and usability study," *JMIR Mental Health*, vol. 6, no. 9, e13887, 2019.

[23] Viknesh Sounderajah et al., "The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence," *Nature Medicine*, Sept. 2025, doi: 10.1038/s41591-025-03953-8.

[24] M. S. Albert et al., "The diagnosis of mild cognitive impairment due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease," *Alzheimer's & Dementia*, vol. 7, no. 3, pp. 270–279, May 2011, doi: 10.1016/j.jalz.2011.03.008.

[25] M. A. Parra, M. Schumacher, and G. Fernández, "A novel peripheral biomarker for mild cognitive impairment and Alzheimer's disease," *Alzheimer's & Dementia*, vol. 16, no. S4, Dec. 2020, doi: 10.1002/alz.044956.

[26] G. S. Collins et al., "TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, Apr. 2024, doi: 10.1136/bmj-2023-078378.

[27] M. D Garrett and R. Valle, "A methodological critique of the National Institute of Aging and Alzheimer's Association Guidelines for Alzheimer's disease, dementia, and mild cognitive impairments," *Dementia*, vol. 15, no. 2, pp. 239–254, Mar. 2014, doi: 10.1177/1471301214525166.